INCH-POUND

MIL-P-45449B (AR)
AMENDMENT 2
06 JUNE 1989
SUPERSEDING
AMENDMENT 1
11 May 1988

MILITARY SPECIFICATION

PALLET, UNITS, WOOD, FOR SHIPMENT OF PROJECTILE METAL PARTS, AND PROJECTILE AMMUNITION

This Amendment forms a part of Military Specification MIL-P-45449B (AR), dated 30 August 1984, and is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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* 3.4.2 Pallet assemblies. Under d (splits or season checks Grade A pallet assemblies only), add the following subparagraph:

"(6) For Type I pallets fabricated with a layer of plywood nailed on top of the cover or base, the defects of this paragraph shall apply only to the lumber beneath the plywood. For the cover assembly, the lumber directly below the plywood shall be considered as part of the cover deckboard and therefore shall meet the requirements of (3) above the second layer of lumber shall be considered the cover cleat and shall meet the requirements of (4) above. For the base assembly, the lumber directly beneath the plywood shall be considered part of the deckboard and shall meet the requirements of (2) above."

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3.8 Preservative treatment. Delete the first and second sentence and substitute:

"Grade A pallets, or the finished wood parts thereof shall be completely immersed for a minimum of one minute with any of the following wood preservatives:

a. PQ 56 reduced with water down to 1.8 percent copper-8-quinolinolate as solution (see 6.4).

b. An emulsion of wood preservative M-GARD W550 (zinc naphthenate) reduced with water down to 3 percent zinc as metal (see 6.5).

c. An emulsion of M-GARD W510 (copper naphthenate) reduced with water down to 2 percent copper as metal (see 6.13).

d. A solution of Cunapsol 5 (copper naphthenate) reduced down with water down to 2 percent copper as metal (see 6.14).

Alternately, Grade A pallets or the finished wood parts thereof shall be completely flooded for a minimum of one minute in PQ 56, M-GARD W550, M-GARD W510 or Cunapsol 5 preservative, as applicable, as to inundate all interior and exterior surfaces (when finished wood parts are dipped)."

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After 3.8.2, add new subparagraph as follows:

"3.8.3 Presence of M-GARD W510 (copper naphthenate) or Cunapsol 5 (copper naphthenate) preservative. When treated with M-GARD W510 or Cunapsol 5, the pallet shall show evidence of discoloration when treated as specified in 4.5.5."

3.9 Pallet identification: After the third sentence ("The letters "PB" shall...."), add sentence as follows:

"The letters "PC" shall be annotated on all pallets subjected to either the Cunapsol 5 (copper naphthenate) or the M-GARD W510 (copper naphthenate) preservative treatment in accordance with 3.8.3."

Fifth sentence: Delete "letters "PA" or "PB" as applicable", and substitute "letters "PA", "PB", or "PC" as applicable".

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4.4.4.3 and 4.4.4.4, Delete in their entirety and substitute:

"4.4.4.3 Presence of wood preservative. (see 3.8), Major defect. A sample of 15 Grade A pallets shall be selected at random for this test. Four members (two cover assemblies and two base assemblies) of the pallets selected shall be subjected to the test specified in 4.5.3, 4.5.4 or 4.5.5 as applicable. If one or more members fail to meet the applicable requirement, the pallet shall be classed defective. Failure of any pallet to comply with the requirement shall be cause for rejection of the lot."

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4.5.4 Presence of M-GARD W550 (zinc naphthenate emulsifiable preservative. Delete in its entirety and substitute:

"4.5.4 Presence of M-GARD W550 (zinc naphthenate emulsifiable) preservative.

4.5.4.1 Materials and equipment. The materials and equipment required are as follows:

a. Reagent. Dissolve 0.1 grams of dithizone (diphenylthiocarbazone) (see 6.10) in either 100 ml of methyl ethyl ketone or 100 ml of chloroform. (Note: This solution may be stored for long periods of time).

b. Eye dropper. An ordinary glass tube eyedropper may be used.

4.5.4.2 Test procedure. Five drops of the solution shall be applied to the wood surface. The indicator will turn red when zinc (M-GARD W550) is present. The color green indicates that the preservative is not present. Both colors fade quickly in the presence of light."

After 4.5.4, add new paragraph as follows:

"4.5.5 Presence of M-GARD W510 (copper naphthenate emulsifiable) or Cunapsol 5 (copper naphthenate) preservative.

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4.5.5.1 Materials and equipment. The materials and equipment required are as follows:

a. Reagent. Dissolve 0.5 grams chrome azurol "S" concentrate (see 6.9) and 5.0 grams sodium acetate in 80 ml of distilled water and then dilute further to 500 ml total with distilled water.

b. Sprayer. A common manual (fly) sprayer type applicator shall be used.

4.5.5.2 Test procedure. Spray solution over surface of dried treated wood. A deep blue color reveals the presence of copper (from copper naphthenate)."

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6.10 Change as follows:

"Dithizone. Dithizone (diphenylthiocarbazone) may be obtained from J. T. Baker Chemical Co., Phillipsburg, New Jersey 08865 or equivalent facility."

Add new paragraphs as follows:

"6.13 M-GARD W510. M-GARD W510 (copper naphthenate), for 2 percent copper as metal, may be obtained from Mooney Chemicals Inc., 2301 Scranton Road, Cleveland, Ohio 44113-9988 or equivalent facility.

6.14 Cunapsol 5. Cunapsol 5 (copper naphthenate), for 2 percent copper as metal, may be obtained from Chapman Chemical Company, P.O. Box 9158, Memphis, TN 38019 or equivalent facility."

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